

The empirical recurrence for OEIS A316119, recovered from its tail

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Abstract

OEIS A316119 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{70} = 3840 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316119 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 0, 2, 3, 5, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 13, 23, 56, 213, 702, 2524, 9034, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jun 24 2018 16:33:46, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	7	c_{19}	−43652	c_{37}	−2137169	c_{55}	−21059587
c_2	−14	c_{20}	−38845	c_{38}	−20726520	c_{56}	54412519
c_3	16	c_{21}	32355	c_{39}	21423307	c_{57}	−61183054
c_4	−58	c_{22}	−130948	c_{40}	−34837293	c_{58}	44715684
c_5	75	c_{23}	113201	c_{41}	49123772	c_{59}	−24066421
c_6	26	c_{24}	−528513	c_{42}	−37440800	c_{60}	9724729
c_7	50	c_{25}	489628	c_{43}	−357731	c_{61}	−1662604
c_8	11	c_{26}	−146751	c_{44}	50076131	c_{62}	−1141628
c_9	−34	c_{27}	858380	c_{45}	−106047483	c_{63}	1497100
c_{10}	−261	c_{28}	−303916	c_{46}	128395321	c_{64}	−1104854
c_{11}	−317	c_{29}	−1039027	c_{47}	−95603680	c_{65}	610200
c_{12}	−409	c_{30}	1724017	c_{48}	54445629	c_{66}	−220340
c_{13}	−1424	c_{31}	−3396645	c_{49}	19381148	c_{67}	6760
c_{14}	2029	c_{32}	3669167	c_{50}	−86559199	c_{68}	40256
c_{15}	−10374	c_{33}	−1429585	c_{51}	141443279	c_{69}	−21056
c_{16}	18848	c_{34}	7847115	c_{52}	−156143689	c_{70}	3840
c_{17}	7552	c_{35}	−3941188	c_{53}	103104041		
c_{18}	46917	c_{36}	2761926	c_{54}	−33992942		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A316119 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \dots - c_{70}$. Its constant term is $-c_{70} = -3840$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 3840.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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